



Linux Foundation

(CKA)

Certified Kubernetes Administrator (CKA)



Be Linux Foundation-CKA Certified Now!

THE **LINUX** FOUNDATION

Certified Kubernetes
Administrator (CKA)

Dumps PDF



Latest Real Exam Q&A, **FIRST ATTEMPT PASS**



81

Q&A.



Full Premium Material PDF – Verified by Experts

✓ Follow us on: [Facebook](#) | [Instagram](#) | [LinkedIn](#) | [reddit](#) | [Twitter](#) | [Quora](#) | [YouTube](#)

Send us your request/inquiry at clearcat.net@gmail.com or connect us for [Live Support](#) any time for **any certification exam dumps pdf** Or for **most asked Interview Q&A PDFs** to ensure your success in first try!!



YouTube.com
/CLEARCATNET



t.Me
/CLEARCATNET

Get any exam latest real exam questions PDF Now-

- ✓ Visit us - WWW.CLEARCATNET.COM
- ✓ Mail us- clearcat.net@gmail.com
- ✓ Live Support- <https://t.me/CLEARCATNET>



We also build professional premium resume by experts to get shortlisted top of profiles

CHAT **t.Me CLEARCATNET** **YouTube.com CLEARCATNET**



Clearcat.net

WWW.CLEARCATNET.COM

Trusted by 56000 Certified Users!

IT Certifications
2500+ Exam & Courses

98.7%
Pass Rate

FIRST ATTEMPT SUCCESS GUARANTEED

Question: 1CKA: Actual Exam Q&A | **CLEARCATNET**

Create a nginx pod with label env=test in engineering namespace See the solution below.

A. Solution:

```
kubectl run nginx --image=nginx --restart=Never --labels=env=test --  
namespace=engineering --dry-run -o yaml > nginx-pod.yaml  
kubectl run nginx --image=nginx --restart=Never --labels=env=test --  
namespace=engineering --dry-run -o yaml | kubectl create -n engineering -f  
YAML File:
```

```
apiVersion: v1  
kind: Pod  
metadata:  
  name: nginx  
  namespace: engineering  
  labels:  
    env: test  
spec:  
  containers:  
  - name: nginx  
    image: nginx
```

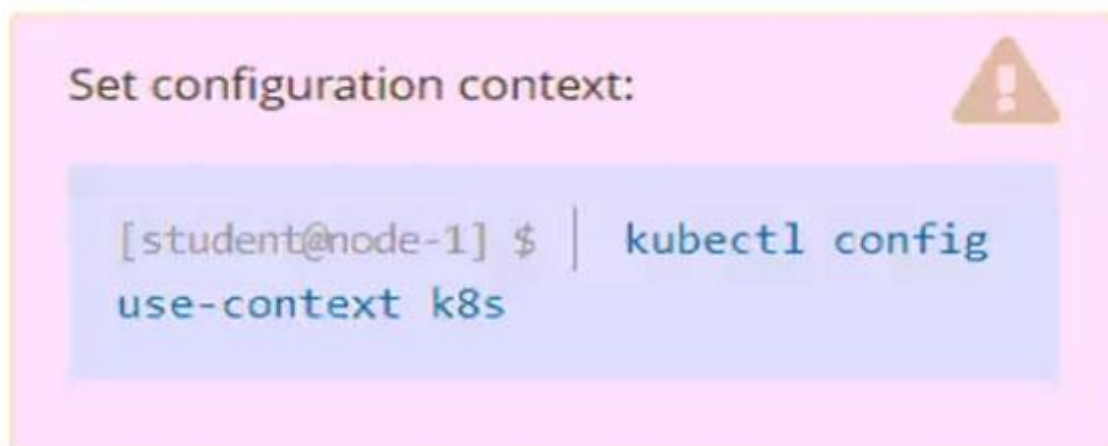
```
imagePullPolicy: IfNotPresent  
restartPolicy: Never
```

```
kubectl create -f nginx-pod.yaml
```

Answer(s): A

Question: 2CKA: Actual Exam Q&A | **CLEARCATNET**

SIMULATION -



Context -

You have been asked to create a new ClusterRole for a deployment pipeline and bind it to a specific ServiceAccount

scoped to a specific namespace.

Task -

Create a new ClusterRole named deployment-clusterrole, which only allows to create the following resource types:

- ⇒ Deployment
- ⇒ Stateful Set
- ⇒ DaemonSet

Create a new ServiceAccount named cicd-token in the existing namespace app-team1.

Bind the new ClusterRole deployment-clusterrole to the new ServiceAccount cicd-token, limited to the namespace app-team1.

Correct

Answer:

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

student@node-1:~$ kubectl config use-context k8s
Switched to context "k8s".
student@node-1:~$ kubectl create clusterrole deployment-clusterrole --verb=create --resource=Deployment,StatefulSet,Daemon
et
clusterrole.rbac.authorization.k8s.io/deployment-clusterrole created
student@node-1:~$ kubectl create sa cicd-token --namespace app-team1
serviceaccount/cicd-token created
student@node-1:~$ kubectl create clusterrolebinding deploy-b --clusterrole=deployment-clusterrole --serviceaccount=app-tea
m1:cicd-token
clusterrolebinding.rbac.authorization.k8s.io/deploy-b created
student@node-1:~$
```

Question: 3

CKA: Actual Exam Q&A | CLEARCATNET

SIMULATION -



Task -

Set the node named ek8s-node-0 as unavailable and reschedule all the pods running on it.

Correct

Answer:

```

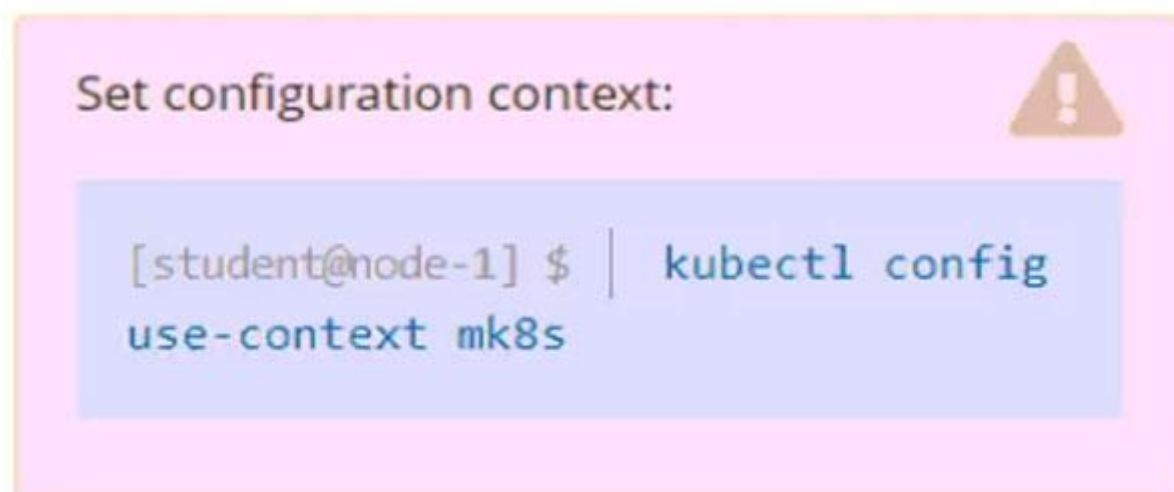
student@node-1:~$ kubectl config use-context ek8s
Switched to context "ek8s".
student@node-1:~$ kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
ek8s-master-0       Ready     control-plane,master   67d   v1.23.1
ek8s-node-0         Ready     <none>    67d   v1.23.1
ek8s-node-1         Ready     <none>    67d   v1.23.1
student@node-1:~$ kubectl drain ek8s-node-1 --ignore-daemonsets
node/ek8s-node-1 cordoned
error: unable to drain node "ek8s-node-1" due to error:cannot delete Pods with local storage (use --delete-emptydir-data to
override): kube-system/metrics-server-7cb5455c67-m6qvq, continuing command...
There are pending nodes to be drained:
  ek8s-node-1
cannot delete Pods with local storage (use --delete-emptydir-data to override): kube-system/metrics-server-7cb5455c67-m6qvq
student@node-1:~$ kubectl drain ek8s-node-1 --ignore-daemonsets --delete-emptydir-data
node/ek8s-node-1 already cordoned
WARNING: ignoring DaemonSet-managed Pods: kube-system/kube-flannel-ds-chvkf, kube-system/kube-proxy-7pf29
evicting pod kube-system/metrics-server-7cb5455c67-m6qvq
evicting pod default/nginx-5cb786cffd-vjbs8
pod/nginx-5cb786cffd-vjbs8 evicted
pod/metrics-server-7cb5455c67-m6qvq evicted
node/ek8s-node-1 drained
student@node-1:~$ kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
ek8s-master-0       Ready     control-plane,master   67d   v1.23.1
ek8s-node-0         Ready     <none>    67d   v1.23.1
ek8s-node-1         Ready,SchedulingDisabled <none>    67d   v1.23.1
student@node-1:~$

```

Question: 4

CKA: Actual Exam Q&A | **CLEARCATNET**

SIMULATION -



Task -

Given an existing Kubernetes cluster running version 1.22.1, upgrade all of the Kubernetes control plane and node components on the master node only to version 1.22.2.

Be sure to drain the master node before upgrading it and uncordon it after the upgrade.

You can `ssh` to the master node using:



```
[student@node-1] $ | ssh mk8s-master-0
```

You can assume elevated privileges on the master node with the following command:

```
[student@mk8s-master-0] $ | sudo -i
```

You are also expected to upgrade kubelet and kubectl on the master node.

Do not upgrade the worker nodes, etcd, the container manager, the CNI plugin, the DNS service or any other addons.



Correct
Answer:


```

student@node-1:~$ kubectl config use-context mk8s
Switched to context "mk8s".
student@node-1:~$ kubectl config use-context mk8s
Switched to context "mk8s".
student@node-1:~$ kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
mk8s-master-0       Ready    control-plane,master   67d   v1.22.1
mk8s-node-0         Ready    <none>    67d   v1.22.1
student@node-1:~$ kubectl drain mk8s-master-0 --ignore-daemonsets
node/mk8s-master-0 cordoned
WARNING: ignoring DaemonSet-managed Pods: kube-system/kube-flannel-ds-jxzm, kube-system/kube-proxy-9rzg9
evicting pod kube-system/coredns-78fcd69978-tt2b8
evicting pod default/nginx-74b46d4cfc-dfkvs
evicting pod kube-system/coredns-78fcd69978-nbkmz
pod/nginx-74b46d4cfc-dfkvs evicted
pod/coredns-78fcd69978-tt2b8 evicted
pod/coredns-78fcd69978-nbkmz evicted
node/mk8s-master-0 drained
student@node-1:~$ kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
mk8s-master-0       Ready,SchedulingDisabled control-plane,master   67d   v1.22.1
mk8s-node-0         Ready    <none>    67d   v1.22.1
student@node-1:~$ ssh mk8s-master-0
Warning: Permanently added '10.250.5.55' (ECDSA) to the list of known hosts.
Welcome to Ubuntu 20.04.3 LTS (GNU/Linux 5.11.0-1028-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Mon Apr 25 09:30:48 UTC 2022

System load: 1.98           Users logged in: 0
Usage of /: 83.2% of 67.79GB IPv4 address for cni0: 10.244.0.1
Memory usage: 2%           IPv4 address for docker0: 172.17.0.1
Swap usage: 0%             IPv4 address for eth0: 10.250.5.55
Processes: 85

30 updates can be applied immediately.
15 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

student@mk8s-master-0:~$ sudo -i
root@mk8s-master-0:~$ apt install kubeadm=1.22.2-00 kubelet=1.22.2-00 kubectl=1.22.2-00
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages will be upgraded:
  kubeadm kubectl kubelet
3 upgraded, 0 newly installed, 0 to remove and 27 not upgraded.
Need to get 39.6 MB of archives.
After this operation, 0 B of additional disk space will be used.
Get:1 https://packages.cloud.google.com/apt/kubernetes-xenial/main amd64 kubelet amd64 1.22.2-00 [21.9 MB]
Get:2 https://packages.cloud.google.com/apt/kubernetes-xenial/main amd64 kubectl amd64 1.22.2-00 [9038 kB]
Get:3 https://packages.cloud.google.com/apt/kubernetes-xenial/main amd64 kubeadm amd64 1.22.2-00 [8718 kB]
Fetched 39.6 MB in 13s (3156 kB/s)
(Reading database ... 33901 files and directories currently installed.)
Preparing to unpack .../kubelet_1.22.2-00_amd64.deb ...
Unpacking kubelet (1.22.2-00) over (1.22.1-00) ...
Preparing to unpack .../kubectl_1.22.2-00_amd64.deb ...
Unpacking kubectl (1.22.2-00) over (1.22.1-00) ...
Preparing to unpack .../kubeadm_1.22.2-00_amd64.deb ...
Unpacking kubeadm (1.22.2-00) over (1.22.1-00) ...
Setting up kubectl (1.22.2-00) ...
Setting up kubelet (1.22.2-00) ...
Setting up kubeadm (1.22.2-00) ...
root@mk8s-master-0:~$ apt install kubeadm=1.22.2-00 kubelet=1.22.2-00 kubectl=1.22.2-00
Reading package lists... Done
Building dependency tree
Reading state information... Done
kubeadm is already the newest version (1.22.2-00).
kubectl is already the newest version (1.22.2-00).
kubelet is already the newest version (1.22.2-00).
0 upgraded, 0 newly installed, 0 to remove and 27 not upgraded.
root@mk8s-master-0:~$ kubeadm upgrade plan
[upgrade/config] Making sure the configuration is correct:
[upgrade/config] Reading configuration from the cluster...
[upgrade/config] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
[preflight] Running pre-flight checks.
[upgrade] Running cluster health checks
[upgrade] Fetching available versions to upgrade to
[upgrade/versions] Cluster version: v1.22.1
[upgrade/versions] kubeadm version: v1.22.2

```


COMPONENT	CURRENT	TARGET
kubelet	1 x v1.22.1	v1.22.9
	1 x v1.22.2	v1.22.9

Upgrade to the latest version in the v1.22 series:

COMPONENT	CURRENT	TARGET
kube-apiserver	v1.22.1	v1.22.9
kube-controller-manager	v1.22.1	v1.22.9
kube-scheduler	v1.22.1	v1.22.9
kube-proxy	v1.22.1	v1.22.9
CoreDNS	v1.8.4	v1.8.4
etcd	3.5.0-0	3.5.0-0

You can now apply the upgrade by executing the following command:

```
kubeadm upgrade apply v1.22.9
```

Note: Before you can perform this upgrade, you have to update kubeadm to v1.22.9.

The table below shows the current state of component configs as understood by this version of kubeadm. Configs that have a "yes" mark in the "MANUAL UPGRADE REQUIRED" column require manual config upgrade or resetting to kubeadm defaults before a successful upgrade can be performed. The version to manually upgrade to is denoted in the "PREFERRED VERSION" column.

API GROUP	CURRENT VERSION	PREFERRED VERSION	MANUAL UPGRADE REQUIRED
kubeproxy.config.k8s.io	v1alpha1	v1alpha1	no
kubelet.config.k8s.io	v1beta1	v1beta1	no

```
root@mk8s-master-0:~# kubeadm upgrade apply v1.22.2
[upgrade/config] Making sure the configuration is correct:
[upgrade/config] Reading configuration from the cluster...
[upgrade/config] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
[preflight] Running pre-flight checks.
[upgrade] Running cluster health checks
[upgrade/version] You have chosen to change the cluster version to "v1.22.2"
[upgrade/versions] Cluster version: v1.22.1
[upgrade/versions] kubeadm version: v1.22.2
[upgrade/confirm] Are you sure you want to proceed with the upgrade? [y/N]: y
[upgrade/prepull] Pulling images required for setting up a Kubernetes cluster
[upgrade/prepull] This might take a minute or two, depending on the speed of your internet connection
[upgrade/prepull] You can also perform this action in beforehand using 'kubeadm config images pull'
[upgrade/apply] Upgrading your Static Pod-hosted control plane to version "v1.22.2"...
Static pod: kube-apiserver-mk8s-master-0 hash: b1d9f2b63ce85cb6310a6d8f6f728f03
Static pod: kube-controller-manager-mk8s-master-0 hash: 91af4173de8872b5f7aec58b2fc0f1fc
Static pod: kube-scheduler-mk8s-master-0 hash: d98fe109788b5b498301dd6c53afcfa9
[upgrade/etcd] Upgrading to TLS for etcd
Static pod: etcd-mk8s-master-0 hash: 6828726f91dba72616d11ac4a737e533
[upgrade/staticpods] Preparing for "etcd" upgrade
[upgrade/staticpods] Current and new manifests of etcd are equal, skipping upgrade
[upgrade/etcd] Waiting for etcd to become available
[upgrade/staticpods] Writing new Static Pod manifests to "/etc/kubernetes/tmp/kubeadm-upgraded-manifests804306747"
[upgrade/staticpods] Preparing for "kube-apiserver" upgrade
[upgrade/staticpods] Renewing apiserver certificate
[upgrade/staticpods] Renewing apiserver-kubelet-client certificate
[upgrade/staticpods] Renewing front-proxy-client certificate
[upgrade/staticpods] Renewing apiserver-etcd-client certificate
[upgrade/staticpods] Moved new manifest to "/etc/kubernetes/manifests/kube-apiserver.yaml" and backed up old manifest to "/etc/kubernetes/tmp/kubeadm-backup-manifests-2022-04-25-15-11-18/kube-apiserver.yaml"
[upgrade/staticpods] Waiting for the kubelet to restart the component
[upgrade/staticpods] This might take a minute or longer depending on the component/version gap (timeout 5m0s)
Static pod: kube-scheduler-mk8s-master-0 hash: d98fe109788b5b498301dd6c53afcfa9
Static pod: kube-scheduler-mk8s-master-0 hash: acb75f76060c8873ac4bf8b2fcl9466
[apiclient] Found 1 Pods for label selector component=kube-scheduler
[upgrade/staticpods] Component "kube-scheduler" upgraded successfully!
[upgrade/postupgrade] Applying label node-role.kubernetes.io/control-plane='' to Nodes with label node-role.kubernetes.io/master= (deprecated)
[upload-config] Storing the configuration used in ConfigMap "kubeadm-config" in the "kube-system" Namespace
[kubelet] Creating a ConfigMap "kubelet-config-1.22" in namespace kube-system with the configuration for the kubelets in the cluster
[kubelet-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[bootstrap-token] configured RBAC rules to allow Node Bootstrap tokens to get nodes
[bootstrap-token] configured RBAC rules to allow Node Bootstrap tokens to post CSRs in order for nodes to get long term certificate credentials
[bootstrap-token] configured RBAC rules to allow the csrapprover controller automatically approve CSRs from a Node Bootstrap Token
[bootstrap-token] configured RBAC rules to allow certificate rotation for all node client certificates in the cluster
[addons] Applied essential addon: CoreDNS
[addons] Applied essential addon: kube-proxy
```

[upgrade/successful] SUCCESS! Your cluster was upgraded to "v1.22.2". Enjoy!

[upgrade/kubelet] Now that your control plane is upgraded, please proceed with upgrading your kubelets if you haven't already done so.

```
root@mk8s-master-0:~# systemctl restart kubelet
```

```
root@mk8s-master-0:~# exit
```

```
logout
```

```
student@mk8s-master-0:~$ exit
```

```
logout
```

Connection to 10.250.5.55 closed.

```
student@node-1:~$ kubectl uncordon mk8s-master-0
```

```
node/mk8s-master-0 uncordoned
```

```
student@node-1:~$ kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
mk8s-master-0	Ready	control-plane,master	67d	v1.22.2

No configuration context change required
for this task.



Ensure, however, that you have returned to the
base node before starting to work on this task:

```
[student@mk8s-master-0] $ | exit
```

Task -

First, create a snapshot of the existing etcd instance running at <https://127.0.0.1:2379>, saving the snapshot to `/var/lib/backup/etcd-snapshot.db`.

The following TLS certificates/key are
supplied for connecting to the server
with `etcdctl`:



- CA certificate: `/opt/KUIN00601/ca.crt`
- Client certificate:
`/opt/KUIN00601/etcd-client.crt`
- Client key:
`/opt/KUIN00601/etcd-client.key`

Creating a snapshot of the given
instance is expected to complete in
seconds.



If the operation seems to hang, something's
likely wrong with your command. Use `CTRL`
`+ C` to cancel the operation and try again.

Next, restore an existing, previous snapshot located at `/var/lib/backup/etcd-snapshot-previous.db`.

**Correct
Answer:**

Correct Answers-

```
student@node-1:~$ kubectl config use-context hk8s
Switched to context "hk8s".
student@node-1:~$ vim policy.yml
student@node-1:~$ kubectl label ns my-app project=my-app
namespace/my-app labeled
student@node-1:~$ kubectl describe ns my-app
Name:          my-app
Labels:        kubernetes.io/metadata.name=my-app
               project=my-app
Annotations:   <none>
Status:        Active

No resource quota.

No LimitRange resource.
student@node-1:~$ kubectl create -f policy.yml
networkpolicy.networking.k8s.io/allow-port-from-namespace created
student@node-1:~$
```

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-port-from-namespace
  namespace: fubar
spec:
  podSelector: {}
  policyTypes:
  - Ingress
  ingress:
  - from:
    - namespaceSelector:
        matchLabels:
          project: my-app
  ports:
  - protocol: TCP
    port: 9000
```

Question: 7

CKA: Actual Exam Q&A | **CLEARCATNET**

SIMULATION -

Set configuration context:



```
[student@node-1] $ | kubectl config
use-context k8s
```

Task -

Reconfigure the existing deployment front-end and add a port specification named http exposing port 80/tcp of the existing container nginx.

Create a new service named front-end-svc exposing the container port http.

Configure the new service to also expose the individual Pods via a NodePort on the nodes on which they are scheduled.

Correct Answer:

```
student@node-1:~$ kubectl config use-context k8s
Switched to context "k8s".
student@node-1:~$ kubectl get deployments.apps
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
front-end     2/2     2            2           5h57m
presentation  2/2     2            2           5h56m
student@node-1:~$ kubectl edit deployments.apps front-end
```

```
# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file will be
# reopened with the relevant failures.
#
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations:
    deployment.kubernetes.io/revision: "1"
    kubernetes.io/last-applied-configuration: |
      {"apiVersion":"apps/v1","kind":"Deployment","metadata":{"annotations":{},"name":"front-end","namespace":"default"},"spec":{"replicas":2,"selector":{"matchLabels":{"app":"front-end"},"template":{"metadata":{"labels":{"app":"front-end"},"spec":{"containers":[{"image":"nginx:1.14.2","name":"nginx"}]}}}}}}
  creationTimestamp: "2022-04-25T09:24:15Z"
  generation: 1
  name: front-end
  namespace: default
  resourceVersion: "3939"
  uid: 1db4fd19-6a6e-4639-a39e-25f836be0017
spec:
  progressDeadlineSeconds: 600
  replicas: 2
  revisionHistoryLimit: 10
  selector:
    matchLabels:
      app: front-end
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: front-end
    spec:
      containers:
      - image: nginx:1.14.2
        imagePullPolicy: IfNotPresent
        name: nginx
        ports:
        - containerPort: 80
          name: http
        resources: {}
        terminationMessagePath: /dev/termination-log
        terminationMessagePolicy: File
      dnsPolicy: ClusterFirst
      restartPolicy: Always
      schedulerName: default-scheduler
      securityContext: {}
      terminationGracePeriodSeconds: 30
status:
  availableReplicas: 2
:~$
```



```

student@node-1:~$ kubectl config use-context k8s
Switched to context "k8s".
student@node-1:~$ kubectl get deployments.apps
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
front-end     2/2     2            2           5h57m
presentation  2/2     2            2           5h56m
student@node-1:~$ kubectl edit deployments.apps front-end
deployment.apps/front-end edited
student@node-1:~$ kubectl expose deployment front-end --name=front-end-svc --port=80 --type=NodePort --protocol=TCP
service/front-end-svc exposed
student@node-1:~$ kubectl describe svc front-end-svc
Name:          front-end-svc
Namespace:     default
Labels:        <none>
Annotations:   <none>
Selector:      app=front-end
Type:          NodePort
IP Family Policy: SingleStack
IP Families:   IPv4
IP:            10.107.66.230
IPs:           10.107.66.230
Port:          <unset> 80/TCP
TargetPort:    80/TCP
NodePort:      <unset> 30392/TCP
Endpoints:     10.244.1.9:80,10.244.2.8:80
Session Affinity: None
External Traffic Policy: Cluster
Events:        <none>
student@node-1:~$ █

```

Question: 8

CKA: Actual Exam Q&A | **CLEARCATNET**

SIMULATION -

Set configuration context:



```
[student@node-1] $ | kubectl config
use-context k8s
```

Task -

Scale the deployment presentation to 3 pods.

Correct Answer:

:

```

student@node-1:~$ kubectl config use-context k8s
Switched to context "k8s".
student@node-1:~$ vim ping.yml
student@node-1:~$ kubectl create -f ping.yml
ingress.networking.k8s.io/ping created
student@node-1:~$ kubectl config use-context k8s
Switched to context "k8s".
student@node-1:~$ kubectl get deployments.apps
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
front-end            2/2     2             2           6h2m
presentation         2/2     2             2           6h1m
student@node-1:~$ kubectl scale deployment presentation --replicas=3
deployment.apps/presentation scaled
student@node-1:~$ kubectl get deployments.apps
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
front-end            2/2     2             2           6h2m
presentation         2/3     3             2           6h1m
student@node-1:~$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
big-corp-app                        1/1     Running   0           5h58m
foo                                 1/1     Running   0           5h58m
front-end-6bc87b9748-n7v8h         1/1     Running   0           3m47s
front-end-6bc87b9748-zmb8g         1/1     Running   0           3m45s
overloaded-cpu-98b9se               1/1     Running   0           5h57m
overloaded-cpu-ab2d3s               1/1     Running   0           5h57m
overloaded-cpu-kipb9a               1/1     Running   0           5h57m
presentation-684cd7ccb6-4gf56       1/1     Running   0           6h1m
presentation-684cd7ccb6-6zjls       1/1     Running   0           13s
presentation-684cd7ccb6-vshxj       1/1     Running   0           6h1m
student@node-1:~$

```

Question: 9

CKA: Actual Exam Q&A | **CLEARCATNET**

SIMULATION -

Set configuration context:



```

[student@node-1] $ | kubectl config
use-context k8s

```

Task -

Schedule a pod as follows:

- ⇒ Name: nginx-kusc00401
- ⇒ Image: nginx
- ⇒ Node selector: disk=ssd

ect

Answer:

```

student@node-1:~$ kubectl config use-context k8s
Switched to context "k8s".
student@node-1:~$ kubectl run nginx-kusc00401 --image=nginx --dry-run -o yaml

```

```

apiVersion: v1
kind: Pod
metadata:
  labels:
    run: nginx-kusc00401
    name: nginx-kusc00401
spec:
  containers:
  - image: nginx
    name: nginx
  nodeSelector:
    disk: ssd

```

```

student@node-1:~$ kubectl config use-context k8s
Switched to context "k8s".
student@node-1:~$ kubectl run nginx-kusc00401 --image=nginx --dry-run -o yaml > n.yaml
W0425 15:27:18.981213 3507450 helpers.go:598] --dry-run is deprecated and can be replaced with --dry-run=client.
student@node-1:~$ vim n.yaml
student@node-1:~$ kubectl create -f n.yaml
pod/nginx-kusc00401 created
student@node-1:~$ kubectl get pods -o wide | grep 401
nginx-kusc00401      1/1      Running    0           12s      10.244.2.10   k8s-node-1   <none>
student@node-1:~$

```

Question: 10

CKA: Actual Exam Q&A | [CLEARCATNET](#)

Create a namespace called 'development' and a pod with image nginx called nginx on this namespace.

A. Solution:

```

kubectl create namespace development
kubectl run nginx --image=nginx --restart=Never -n development

```

Answer(s): A

Question: 11

CKA: Actual Exam Q&A | [CLEARCATNET](#)

Get list of all pods in all namespaces and write it to file "/opt/pods-list.yaml"

A. Solution:

```

kubectl get po all-namespaces > /opt/pods-list.yaml

```

Answer(s): A

Question: 12

CKA: Actual Exam Q&A | [CLEARCATNET](#)

Create a pod with image nginx called nginx and allow traffic on port 80

A. Solution:

```

kubectl run nginx --image=nginx --restart=Never --port=80

```

Answer(s): A

Question: 13CKA: Actual Exam Q&A | **CLEARCATNET**

Create a busybox pod that runs the command "env" and save the output to "envpod" file

A. Solution:

```
kubectrl run busybox --image=busybox --restart=Never -rm -it -- env > envpod.yaml
```

Answer(s): A

Question: 14CKA: Actual Exam Q&A | **CLEARCATNET**

List pod logs named "frontend" and search for the pattern "started" and write it to a file "/opt/error-logs"

A. Solution:

```
Kubectrl logs frontend | grep -i "started" > /opt/error-logs
```

Answer(s): A

Question: 15CKA: Actual Exam Q&A | **CLEARCATNET**

Create a pod that echo "hello world" and then exists. Have the pod deleted automatically when it's completed

A. Solution:

```
kubectrl run busybox --image=busybox -it --rm --restart=Never -- /bin/sh -c 'echo hello world'
kubectrl get po # You shouldn't see pod with the name "busybox"
```

Answer(s): A

Question: 16CKA: Actual Exam Q&A | **CLEARCATNET**

Create a pod with environment variables as var1=value1. Check the environment variable in pod

A. Solution:

```
kubectrl run nginx --image=nginx --restart=Never --env=var1=value1 # then
kubectrl exec -it nginx -- env
# or
kubectrl exec -it nginx -- sh -c 'echo $var1'
# or
kubectrl describe po nginx | grep value1
```

Answer(s): A

Question: 17CKA: Actual Exam Q&A | **CLEARCATNET**

Get list of all the pods showing name and namespace with a jsonpath expression.

A. Solution:

```
kubectl get pods -o=jsonpath="{.items[*]['metadata.name' , 'metadata.namespace']}"
```

Answer(s): A

Question: 18CKA: Actual Exam Q&A | **CLEARCATNET**

Check the image version in pod without the describe command

A. Solution:

```
kubectl get po nginx -o  
jsonpath='{.spec.containers[].image}'
```

Answer(s): A

Question: 19CKA: Actual Exam Q&A | **CLEARCATNET**

List the nginx pod with custom columns POD_NAME and POD_STATUS

A. Solution:

```
kubectl get po -o=custom-columns="POD_NAME:.metadata.name,  
POD_STATUS:.status.containerStatuses[].state"
```

Answer(s): A

List all the pods sorted by name

A. Solution:

```
kubect1 get pods --sort-by=.metadata.name
```

Answer(s): A

Create a pod that having 3 containers in it? (Multi-Container)

A. Solution:

```
image=nginx, image=redis, image=consul
Name nginx container as "nginx-container"
Name redis container as "redis-container"
Name consul container as "consul-container"
Create a pod manifest file for a container and append container section for rest of the
images
kubectl run multi-container --generator=run-pod/v1 --image=nginx -- dry-run -o yaml >
multi-container.yaml
# then
vim multi-container.yaml
apiVersion: v1
kind: Pod
metadata:

labels:
run: multi-container
name: multi-container
spec:
containers:
- image: nginx
name: nginx-container
- image: redis
name: redis-container
- image: consul
name: consul-container
restartPolicy: Always
```

Answer(s): A

Create 2 nginx image pods in which one of them is labelled with env=prod and another one labelled with env=dev and verify the same.

A. Solution:

```
kubectrl run --generator=run-pod/v1 --image=nginx -- labels=env=prod nginx-prod --dry-run -
o yaml > nginx-prodpod.yaml Now, edit nginx-prod-pod.yaml file and remove entries like
"creationTimestamp:
null" "dnsPolicy: ClusterFirst"
vim nginx-prod-pod.yaml
apiVersion: v1
kind: Pod
metadata:
labels:
env: prod
name: nginx-prod
spec:
containers:
- image: nginx
name: nginx-prod
restartPolicy: Always
# kubectrl create -f nginx-prod-pod.yaml
kubectrl run --generator=run-pod/v1 --image=nginx --
labels=env=dev nginx-dev --dry-run -o yaml > nginx-dev-pod.yaml
apiVersion: v1
kind: Pod
metadata:
labels:
env: dev
name: nginx-dev
spec:

containers:
- image: nginx
name: nginx-dev
restartPolicy: Always
# kubectrl create -f nginx-prod-dev.yaml
Verify :
kubectrl get po --show-labels
kubectrl get po -l env=prod
kubectrl get po -l env=dev
```

Answer(s): A

A. Solution:

```
Kubect1 get po -o wide
Using JsonPath
kubect1 get pods -o=jsonpath='{range
.items[*]}{.metadata.name}{"\t"}{.status.podIP}{"\n"}{end}'
```

Answer(s): A

Question: 23

CKA: Actual Exam Q&A | CLEARCATNET

Print pod name and start time to "/opt/pod-status" file

A. Solution:

```
kubect1 get pods -o=jsonpath='{range
.items[*]}{.metadata.name}{"\t"}{.status.podIP}{"\n"}{end}'
```

Answer(s): A

Question: 24

CKA: Actual Exam Q&A | CLEARCATNET

Check the Image version of nginx-dev pod using jsonpath

A. Solution:

```
kubect1 get po nginx-dev -o
jsonpath='{.spec.containers[].image}{"\n"}'
```

Answer(s): A

Question: 25

CKA: Actual Exam Q&A | CLEARCATNET

Create a busybox pod and add "sleep 3600" command

A. Solution:

```
kubectl run busybox --image=busybox --restart=Never -- /bin/sh -c "sleep 3600"
```

Answer(s): A

Question: 26

CKA: Actual Exam Q&A | CLEARCATNET

Create an nginx pod and list the pod with different levels of verbosity

A. Solution:

```
// create a pod
kubectl run nginx --image=nginx --restart=Never --port=80 // List the pod with different
verbosity
kubectl get po nginx --v=7
```



```
kubectl get po nginx --v=8  
kubectl get po nginx --v=9
```

Answer(s): A

Question: 27

CKA: Actual Exam Q&A | CLEARCATNET

List the nginx pod with custom columns POD_NAME and POD_STATUS

A. Solution:

```
kubectl get po -o=custom-columns="POD_NAME:.metadata.name,
```

```
POD_STATUS:.status.containerStatuses[].state"
```

Answer(s): A

Question: 28

CKA: Actual Exam Q&A | CLEARCATNET

List all the pods sorted by name

A. Solution:

```
kubectl get pods --sort-by=.metadata.name
```

Answer(s): A

Question: 29

CKA: Actual Exam Q&A | CLEARCATNET

List all the pods sorted by created timestamp

A. Solution:

```
kubect1 get pods--sort-by=.metadata.creationTimestamp
```

Question: 30

CKA: Actual Exam Q&A | CLEARCATNET

List all the pods showing name and namespace with a json path expression

A. Solution:

```
kubectl get pods -o=jsonpath="{.items[*]['metadata.name', 'metadata.namespace']}"
```

Answer(s): A

Question: 31

CKA: Actual Exam Q&A | CLEARCATNET

List "nginx-dev" and "nginx-prod" pod and delete those pods

A. Solution:

```
kubect1 get pods -o wide  
kubectl delete po "nginx-dev"  
kubectl delete po "nginx-prod"
```

Answer(s): A

Question: 32

CKA: Actual Exam Q&A | CLEARCATNET

Delete the pod without any delay (force delete)

A. Solution:

kubect1 delete po "POD-NAME" --grace-period=0 force

Answer(s): A

Question: 33

CKA: Actual Exam Q&A | CLEARCATNET

Create a redis pod and expose it on port 6379

A. Solution:

kubect1 run redis --image=redis --restart=Never --port=6379

YAML File :

apiVersion: v1

kind: Pod

metadata:

labels:

run: redis

name: redis

spec:

containers:

- image: redis

name: redis

ports:

- containerPort: 6379

RestartPolicy: Always

Answer(s): A

Question: 34

CKA: Actual Exam Q&A | CLEARCATNET

Create the nginx pod with version 1.17.4 and expose it on port 80

A. Solution:

kubect1 run nginx --image=nginx:1.17.4 --restart=Never -- port=80

Answer(s): A

Question: 35

CKA: Actual Exam Q&A | CLEARCATNET

Change the Image version to 1.15-alpine for the pod you just created and verify the image version is updated.

A. Solution:

```
kubect1 set image pod/nginx nginx=nginx:1.15-alpine
kubect1 describe po nginx
// another way it will open vi editor and change the version kubectl edit po nginx
kubect1 describe po nginx
```

Answer(s): A

Question: 36

CKA: Actual Exam Q&A | CLEARCATNET

Change the Image version back to 1.17.1 for the pod you just updated and observe the changes

A. Solution:

```
kubectl set image pod/nginx nginx=nginx:1.17.1
kubectl describe po nginx
kubectl get po nginx -w # watch it
```

Answer(s): A

Question: 37

CKA: Actual Exam Q&A | CLEARCATNET

Create a redis pod, and have it use a non-persistent storage Note: In exam, you will have access to kubernetes.io site, Refer :

<https://kubernetes.io/docs/tasks/configure-pod-container/configurevolume-storage/>

A. Solution:

```
apiVersion: v1
kind: Pod
metadata:
  name: redis
spec:
  containers:
  - name: redis
    image: redis
    volumeMounts:
    - name: redis-storage
      mountPath: /data/redis
  ports:
  - containerPort: 6379
  volumes:
  - name: redis-storage
    emptyDir: {}
```

Answer(s): A

Question: 38

CKA: Actual Exam Q&A | CLEARCATNET

Create a Pod with three busy box containers with commands "ls; sleep 3600;", "echo Hello World; sleep 3600;" and "echo this is the third container; sleep 3600" respectively and check the status

A. Solution:

```
// first create single container pod with dry run flag kubectl run busybox --image=busybox
--restart=Always --dry-run -o yaml -- bin/sh -c "sleep 3600; ls" > multi-container.yaml // edit
the pod to following yaml and create it
apiVersion: v1
kind: Pod
metadata:
labels:
run: busybox
name: busybox
spec:
containers:
- args:
- bin/sh
- -c
- ls; sleep 3600
image: busybox
name: busybox-container-1
- args:
- bin/sh
- -c
- echo Hello world; sleep 3600
image: busybox
name: busybox-container-2
- args:
- bin/sh
- -c
- echo this is third container; sleep 3600
image: busybox
name: busybox-container-3
restartPolicy: Always
// Verify
Kubectl get pods
```

Answer(s): A

Question: 39

CKA: Actual Exam Q&A | CLEARCATNET

Check logs of each container that "busyboxpod-{1,2,3}"

A. Solution:

```
kubectl logs busybox -c busybox-container-1
kubectl logs busybox -c busybox-container-2
kubectl logs busybox -c busybox-container-3
```

Answer(s): A

Question: 40

CKA: Actual Exam Q&A | CLEARCATNET

Create a Pod with main container busybox and which executes this "while true; do echo `Hi I am from Main container` >> /var/log/index.html; sleep 5; done" and with sidecar container with nginx image which exposes on port 80. Use emptyDir Volume and mount this volume on path /var/log for busybox and on path /usr/share/nginx/html for nginx container. Verify both containers are running.

A. Solution:

```
// create an initial yaml file with this
kubectl run multi-cont-pod --image=busybox --restart=Never -- dry-run -o yaml > multi-
container.yaml
// edit the yml as below and create it
kubectl create -f multi-container.yaml
vim multi-container.yaml
apiVersion: v1
kind: Pod
metadata:
labels:
run: multi-cont-pod
name: multi-cont-pod
spec:
volumes:
- name: var-logs
emptyDir: {}
containers:
- image: busybox
command: ["/bin/sh"]
args: ["-c", "while true; do echo 'Hi I am from Main container' >> /var/log/index.html; sleep
5;done"]
name: main-container
volumeMounts:
- name: var-logs
mountPath: /var/log
- image: nginx
name: sidecar-container
ports:
```



```
- containerPort: 80
volumeMounts:
- name: var-logs
```

```
mountPath: /usr/share/nginx/html
restartPolicy: Never
// Create Pod
kubectl apply -f multi-container.yaml
//Verify
kubectl get pods
```

Answer(s): A

Question: 41 [CKA: Actual Exam Q&A | CLEARCATNET](#)

Exec into both containers and verify that main.txt exist and

A. Solution:
NEED TO WRITE ANSWER FOR THIS

Answer(s): A

Question: 42 [CKA: Actual Exam Q&A | CLEARCATNET](#)

Create an nginx pod and set an env value as 'var1=val1'. Check the env value existence within the pod

A. Solution:

```
kubectl run nginx --image=nginx --restart=Never --env=var1=val1 # then
kubectl exec -it nginx -- env
# or
kubectl exec -it nginx -- sh -c 'echo $var1'
# or
kubectl describe po nginx | grep val1
# or
kubectl run nginx --restart=Never --image=nginx --env=var1=val1 -it --rm env
```

Answer(s): A

Question: 43 [CKA: Actual Exam Q&A | CLEARCATNET](#)

Create a pod with init container which create a file "test.txt" in "workdir" directory. Main container should check a file "test.txt" exists and execute sleep 9999 if the file exists.

A. Solution:

```
// create an initial yaml file with this
```

```
kubectrl run init-cont-pod --image=alpine --restart=Never --dry-run -o
```

```
yaml > init-cont-pod.yaml
```

```
// edit the yml as below and create it
```

```
vim init-cont-pod.yaml
```

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
name: init-cont-pod
```

```
labels:
```

```
app: myapp
```

```
spec:
```

```
volumes:
```

```
- name: test-volume
```

```
emptyDir: {}
```

```
containers:
```

```
- name: main-container
```

```
image: busybox:1.28
```

```
command: ['sh', '-c', 'if [ -f /workdir/test.txt ]; then sleep 9999; fi']
```

```
volumeMounts:
```

```
- name: test-volume
```

```
mountPath: /workdir
```

```
initContainers:
```

```
- name: init-myservice
```

```
image: busybox:1.28
```

```
command: ['sh', '-c', "mkdir /workdir; echo > /workdir/test.txt"]
```

```
volumeMounts:
```

```
- name: test-volume
```

```
mountPath: /workdir
```

```
// Create the pod
```

```
kubectrl apply -f init-cont-pod.yaml
```

```
kubectrl get pods
```

```
// Check Events by doing
```

```
kubectrl describe po init-cont-pod
```

```
Init Containers:
```

```
init-myservice:
```

```
Container ID:
```

```
docker://ebdbf5fad1c95111d9b0e0e2e743c2e347c81b8d4eb5abcccdfe1dd74524 0d4f
```

```
Image: busybox:1.28
```

```
Image ID:
```

```
dockerpullable://busybox@sha256:141c253bc4c3fd0a201d32dc1f493bcf3fff003b6df  
416dea4f41046e0f37d47
```

```
Port: <none>
```

```
Host Port: <none>
```

```
Command:
```

```
sh
-c
mkdir /workdir; echo > /workdir/test.txt
```

State: Terminated
Reason: Completed

Answer(s): A

Question: 44

CKA: Actual Exam Q&A | CLEARCATNET

Create a pod with init container which waits for a service called "myservice" to be created. Once init container completes, the myapp-container should start and print a message "The app is running" and sleep for 3600 seconds.

A. Solution:

```
vim multi-container-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: myapp-pod
  labels:
    app: myapp
spec:
  containers:
  - name: myapp-container
    image: busybox:1.28
    command: ['sh', '-c', 'echo The app is running! && sleep 3600']
  initContainers:
  - name: init-myservice
    image: busybox:1.28
    command: ['sh', '-c', "until nslookup myservice.$(cat
/var/run/secrets/kubernetes.io/serviceaccount/namespace).svc.cluster.local; do echo
waiting for myservice; sleep 2; done"]
    // Check whether service called "myservice" exists
    kubectl get svc
    Note: Pod will not start if service called "myservice" doesn't exist.
    // Now, Create the pod
    kubectl apply -f multi-container-pod.yaml
```

Answer(s): A

Question: 45

CKA: Actual Exam Q&A | CLEARCATNET

Create 5 nginx pods in which two of them is labeled env=prod and three of them is labeled env=dev

A. Solution:

```
kubectrl run nginx-dev1 --image=nginx --restart=Never -- labels=env=dev
kubectrl run nginx-dev2 --image=nginx --restart=Never -- labels=env=dev
kubectrl run nginx-dev3 --image=nginx --restart=Never -- labels=env=dev
kubectrl run nginx-prod1 --image=nginx --restart=Never -- labels=env=prod
kubectrl run nginx-prod2 --image=nginx --restart=Never -- labels=env=prod
```

Answer(s): A

Question: 46

CKA: Actual Exam Q&A | **CLEARCATNET**

Get the pods with label env=dev and output the labels

A. Solution:

```
kubectrl get pods -l env=dev --show-labels
```

Answer(s): A

Question: 47

CKA: Actual Exam Q&A | **CLEARCATNET**

Get the pods with labels env=dev and env=prod and output the labels as well

A. Solution:

```
kubectrl get pods -l 'env in (dev,prod)' --show-labels
```

Answer(s): A

Question: 48

CKA: Actual Exam Q&A | **CLEARCATNET**

Get all the pods with label "env"

A. Solution:

```
kubectrl get pods -L env
```

Answer(s): A

Question: 49

CKA: Actual Exam Q&A | **CLEARCATNET**

Change the label for one of the pod to env=uat and list all the pods to verify

A. Solution:

```
kubectrl label pod/nginx-dev3 env=uat --overwrite
kubectrl get pods --show-labels
```

Answer(s): A

Get list of all the nodes with labels

A. Solution:

```
kubectl get nodes --show-labels
```

Answer(s): A

Create a nginx pod that will be deployed to node with the label "gpu=true"

A. Solution:

```
kubectl run nginx --image=nginx --restart=Always --dry-run -o yaml > nodeselector-  
pod.yaml  
// add the nodeSelector like below and create the pod kubectl apply -f nodeselector-  
pod.yaml  
vim nodeselector-pod.yaml  
apiVersion: v1  
kind: Pod  
metadata:  
  name: nginx  
spec:  
  nodeSelector:  
    gpu: true  
  containers:  
  - image: nginx  
    name: nginx  
    restartPolicy: Always  
kubectl apply -f nodeselector-pod.yaml  
//Verify  
kubectl get po --show-labels  
kubectl get po  
kubectl describe po nginx | grep Node-Selectors
```

Answer(s): A

Annotate the pod with name=webapp

A. Solution:

```
kubectl annotate pod nginx-dev-pod name=webapp  
kubectl annotate pod nginx-prod-pod name=webapp  
// Verify
```



```
kubectl describe po nginx-dev-pod | grep -i annotations kubectl describe po nginx-prod-  
pod | grep -i annotations
```

Answer(s): A

Question: 53

CKA: Actual Exam Q&A | CLEARCATNET

Create a deployment called webapp with image nginx having 5 replicas in it, put the file in /tmp directory with named webapp.yaml

A. Solution:

```
//Create a file using dry run command  
kubectl create deploy --image=nginx --dry-run -o yaml > /tmp/webapp.yaml  
// Now, edit file webapp.yaml and update replicas=5  
apiVersion: apps/v1  
kind: Deployment  
metadata:  
labels:  
app: webapp  
name: webapp  
spec:  
replicas: 5  
selector:  
matchLabels:  
app: webapp  
template:  
metadata:  
labels:  
app: webapp  
spec:  
containers:  
- image: nginx  
name: nginx  
Note: Search "deployment" in kubernetes.io site , you will get the page  
https://kubernetes.io/docs/concepts/workloads/controllers/deplo
```

```
yment/  
// Verify the Deployment  
kubectl get deploy webapp --show-labels  
// Output the YAML file of the deployment webapp  
kubectl get deploy webapp -o yaml
```

Answer(s): A

Question: 54

CKA: Actual Exam Q&A | CLEARCATNET

Get the list of pods of webapp deployment

A. Solution:

```
// Get the label of the deployment
kubectl get deploy --show-labels
// Get the pods with that label
kubectl get pods -l app=webapp
```

Answer(s): A

Question: 55

CKA: Actual Exam Q&A | CLEARCATNET

Scale the deployment from 5 replicas to 20 replicas and verify

A. Solution:

```
kubectl scale deploy webapp --replicas=20
kubectl get deploy webapp
kubectl get po -l app=webapp
```

Answer(s): A

Question: 56

CKA: Actual Exam Q&A | CLEARCATNET

Get the deployment rollout status

A. Solution:

```
kubectl rollout status deploy webapp
```

Answer(s): A

Question: 57

CKA: Actual Exam Q&A | CLEARCATNET

Deployment

Create a deployment of webapp with image nginx:1.17.1 with container port 80 and verify the image version

A. Solution:

```
// Create initial YAML file with dry-run option
kubectl create deploy webapp --image=nginx:1.17.1 --dryrun=client -o yaml > webapp.yaml
vim webapp.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
labels:
app: webapp
name: webapp
spec:
replicas: 1
selector:
```

```
matchLabels:
app: webapp
template:
metadata:
labels:
app: webapp
spec:
containers:
- image: nginx:1.17.1
name: nginx
kubectl create -f webapp.yaml record=true
//Verify Image Version
kubectl describe deploy webapp | grep -i "Image"
Using JsonPath
kubectl get deploy -o=jsonpath='{range.items [*]}{.[*]}
{.metadata.name}{"\t"}{.spec.template.spec.containers[*].i mage}{"\n"}'
```

Answer(s): A

Question: 58

CKA: Actual Exam Q&A | CLEARCATNET

Update the deployment with the image version 1.17.4 and verify

A. Solution:

```
kubectl set image deploy/webapp nginx=nginx:1.17.4
//Verify
kubectl describe deploy webapp | grep Image
kubectl get deploy -o=jsonpath='{range.items [*]}{.[*]}
{.metadata.name}{"\t"}{.spec.template.spec.containers[*].i mage}{"\n"}'
```

Answer(s): A

Question: 59

CKA: Actual Exam Q&A | CLEARCATNET

Check the rollout history and make sure everything is ok after the update

A. Solution:

```
kubectl rollout history deploy webapp
kubectl get deploy webapp --show-labels
kubectl get rs -l app=webapp
kubectl get po -l app=webapp
```

Answer(s): A

Question: 60

CKA: Actual Exam Q&A | CLEARCATNET

Undo the deployment to the previous version 1.17.1 and verify Image has the previous version

A. Solution:

```
kubectl rollout undo deploy webapp  
kubectl describe deploy webapp | grep Image
```

Answer(s): A

Question: 61

CKA: Actual Exam Q&A | **CLEARCATNET**

Update the deployment with the image version 1.16.1 and verify the image and check the rollout history

A. Solution:

```
kubectl set image deploy/webapp nginx=nginx:1.16.1  
kubectl describe deploy webapp | grep Image  
kubectl rollout history deploy webapp
```

Answer(s): A

Question: 62

CKA: Actual Exam Q&A | **CLEARCATNET**

Check the history of deployment

A. Solution:

```
kubectl rollout history deployment webapp
```

Answer(s): A

Question: 63

CKA: Actual Exam Q&A | **CLEARCATNET**

Undo the deployment with the previous version and verify everything is Ok

A. Solution:

```
kubectl rollout undo deploy webapp  
kubectl rollout status deploy webapp  
kubectl get pods
```

Answer(s): A

Question: 64

CKA: Actual Exam Q&A | **CLEARCATNET**

Undo/Rollback deployment to specific revision "1"

A. Solution:

```
// Check Deployment History
kubectl rollout history deployment webapp
//Rollback to particular revision
kubectl rollout undo deployment webapp --to-revision=1
```

Answer(s): A

Question: 65

CKA: Actual Exam Q&A | CLEARCATNET

Check the history of the specific revision of that deployment

A. Solution:

```
kubectl rollout history deployment webapp --revision=3
```

Answer(s): A

Question: 66

CKA: Actual Exam Q&A | CLEARCATNET

Pause the rollout of the deployment

A. Solution:

```
kubectl rollout pause deployment webapp
```

Answer(s): A

Question: 67

CKA: Actual Exam Q&A | CLEARCATNET

Resume the rollout of the deployment

A. Solution:

```
kubectl rollout resume deployment webapp
```

Answer(s): A

Question: 68

CKA: Actual Exam Q&A | CLEARCATNET

Scale the deployment to 5 replicas

A. Solution:

```
kubectl scale deployment webapp replicas=5
//Verify
kubectl get deployment
kubectl get pods
```

Answer(s): A

Question: 69

CKA: Actual Exam Q&A | CLEARCATNET

Scale down the deployment to 1 replica

A. Solution:

```
kubectl scale deployment webapp replicas=1
//Verify
kubectl get deploy
kubectl get po,rs
```

Answer(s): A

Question: 70

CKA: Actual Exam Q&A | CLEARCATNET

Apply the autoscaling to this deployment with minimum 10 and maximum 20 replicas and target CPU of 85% and verify hpa is created and replicas are increased to 10 from 1

A. Solution:

```
kubectl autoscale deploy webapp --min=10 --max=20 --cpu percent=85
kubectl get hpa

kubectl get pod -l app=webapp
```

Answer(s): A

Question: 71

CKA: Actual Exam Q&A | CLEARCATNET

Clean the cluster by deleting deployment and hpa you just Created

A. Solution:

```
kubectl delete deploy webapp
kubectl delete hpa webapp
```

Answer(s): A

Question: 72

CKA: Actual Exam Q&A | CLEARCATNET

Create a pod with environment variables as var1=value1. Check the environment variable in pod

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:


```
kubectl run nginx --image=nginx --restart=Never --env=var1=value1# then
```

```
kubectl exec -it nginx -- env # or  
kubectl exec -it nginx -- sh -c 'echo $var1' # or  
kubectl describe po nginx | grep value1
```

Question: 73

CKA: Actual Exam Q&A | **CLEARCATNET**

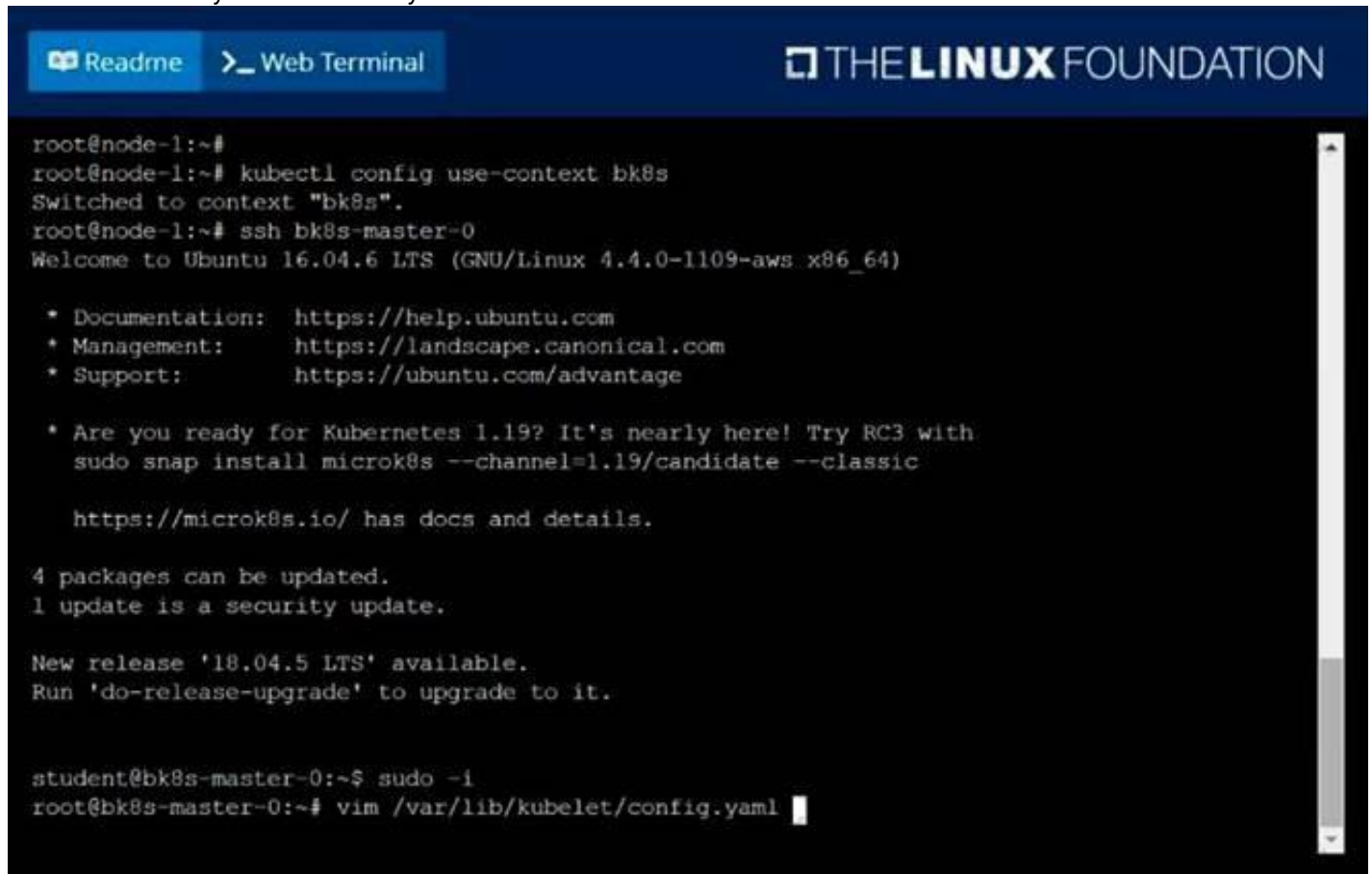
Given a partially-functioning Kubernetes cluster, identify symptoms of failure on the cluster. Determine the node, the failing service, and take actions to bring up the failed service and restore the health of the cluster. Ensure that any changes are made permanently. You can ssh to the relevant nodes (bk8s-master-0 or bk8s-node-0) using: [student@node-1] \$ ssh <nodename> You can assume elevated privileges on any node in the cluster with the following command: [student@nodename] \$ | sudo ?Ci

A. Mastered

B. Not Mastered **Answer: A Explanation:**

solution

F:\Work\Data Entry Work\Data Entry\20200827\CKA\23 C.JPG



The screenshot shows a terminal window with a dark background. At the top, there is a blue header bar with the text 'THE LINUX FOUNDATION' on the right and two buttons on the left: 'Readme' and 'Web Terminal'. The terminal content shows a user at 'root@node-1:~#'. They run 'kubectl config use-context bk8s', which switches the context to 'bk8s'. Then they run 'ssh bk8s-master-0', connecting to the master node. The prompt changes to 'root@bk8s-master-0:~#'. The terminal shows the Ubuntu 16.04.6 LTS login banner, followed by links for documentation, management, and support. It also shows a message about Kubernetes 1.19 and a security update. Finally, the user runs 'sudo -i' to become root, and the prompt changes to 'root@bk8s-master-0:~#'. They then run 'vim /var/lib/kubelet/config.yaml'.

```
root@node-1:~#  
root@node-1:~# kubectl config use-context bk8s  
Switched to context "bk8s".  
root@node-1:~# ssh bk8s-master-0  
Welcome to Ubuntu 16.04.6 LTS (GNU/Linux 4.4.0-1109-aws x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/advantage  
  
* Are you ready for Kubernetes 1.19? It's nearly here! Try RC3 with  
  sudo snap install microk8s --channel=1.19/candidate --classic  
  
https://microk8s.io/ has docs and details.  
  
4 packages can be updated.  
1 update is a security update.  
  
New release '18.04.5 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
student@bk8s-master-0:~$ sudo -i  
root@bk8s-master-0:~# vim /var/lib/kubelet/config.yaml
```

F:\Work\Data Entry Work\Data Entry\20200827\CKA\23 D.JPG

```
authorization:
  mode: Webhook
  webhook:
    cacheAuthorizedTTL: 0s
    cacheUnauthorizedTTL: 0s
clusterDNS:
- 10.96.0.10
clusterDomain: cluster.local
cpuManagerReconcilePeriod: 0s
evictionPressureTransitionPeriod: 0s
fileCheckFrequency: 0s
healthzBindAddress: 127.0.0.1
healthzPort: 10248
httpCheckFrequency: 0s
imageMinimumGCAge: 0s
kind: KubeletConfiguration
nodeStatusReportFrequency: 0s
nodeStatusUpdateFrequency: 0s
rotateCertificates: true
runtimeRequestTimeout: 0s
staticPodPath: /etc/kubernetes/manifests
streamingConnectionIdleTimeout: 0s
syncFrequency: 0s
volumeStatsAggPeriod: 0s
:wg
```

F:\Work\Data Entry Work\Data Entry\20200827\CKA\23 E.JPG

<https://microk8s.io/> has docs and details.

4 packages can be updated.
1 update is a security update.

New release '18.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

```
student@bk8s-master-0:~$ sudo -i
root@bk8s-master-0:~# vim /var/lib/kubelet/config.yaml
root@bk8s-master-0:~# systemctl restart kubelet
root@bk8s-master-0:~# systemctl enable kubelet
root@bk8s-master-0:~# kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
bk8s-master-0	Ready	master	77d	v1.18.2
bk8s-node-0	Ready	<none>	77d	v1.18.2

```
root@bk8s-master-0:~#
root@bk8s-master-0:~# exit
logout
student@bk8s-master-0:~$ exit
logout
Connection to 10.250.4.77 closed.
root@node-1:~#
```

Question: 74**CKA: Actual Exam Q&A | CLEARCATNET**

Create a deployment as follows: Name: nginx-app

➤ Using container nginx with version 1.11.10-alpine. The deployment should contain 3 replicas

Next, deploy the application with new version 1.11.13-alpine, by performing a rolling update.

Finally, rollback that update to the previous version 1.11.10-alpine.

A. Mastered

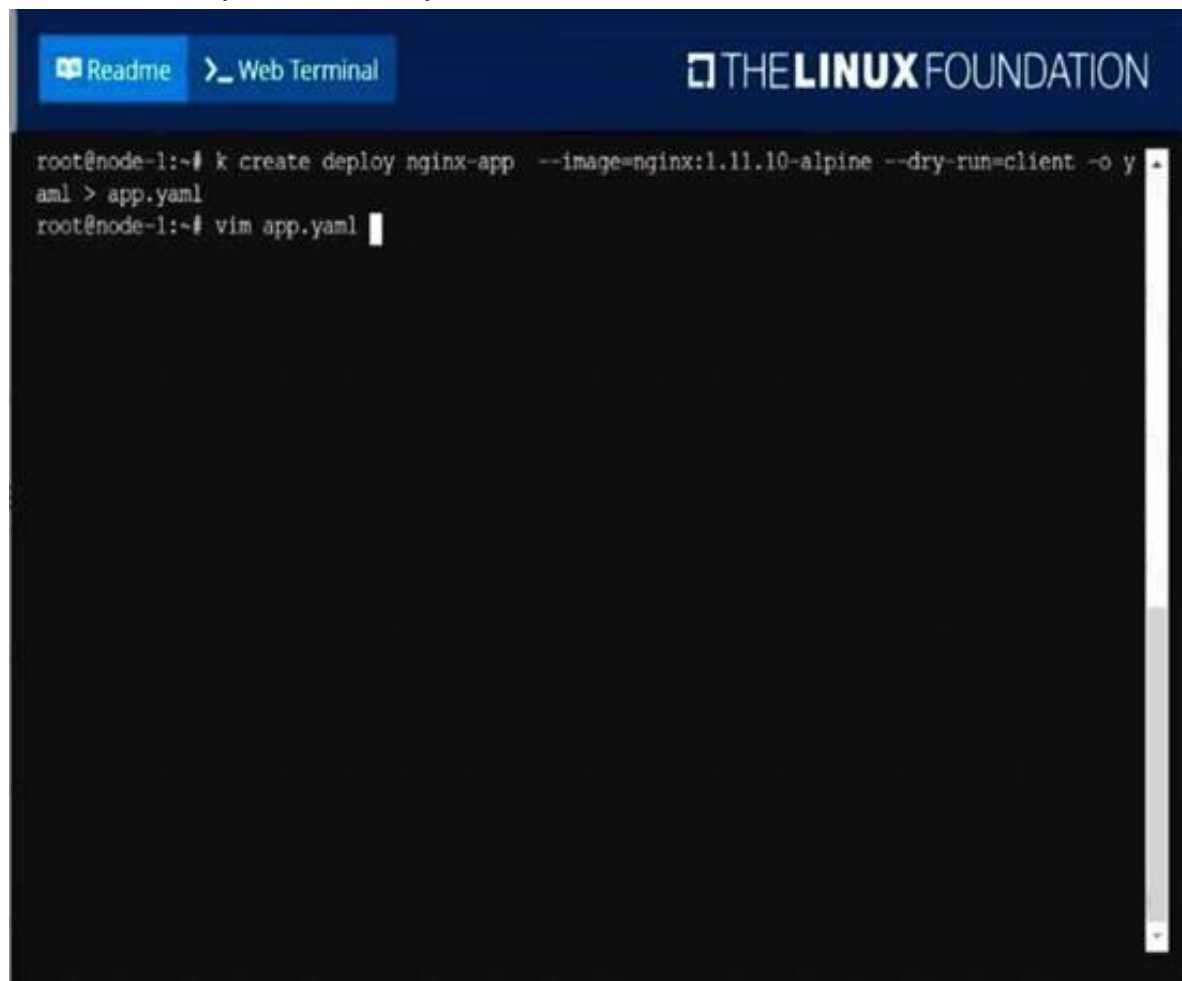
B. Not Mastered

Answer: A

Explanation:

solution

F:\Work\Data Entry Work\Data Entry\20200827\CKA\7 B.JPG



The screenshot shows a web terminal interface with a dark blue header. On the left, there are two buttons: 'Readme' and 'Web Terminal'. On the right, the 'THE LINUX FOUNDATION' logo is displayed. The terminal window shows a root user at a node-1 prompt. The user enters the command 'k create deploy nginx-app --image=nginx:1.11.10-alpine --dry-run=client -o yaml > app.yaml'. The prompt changes to 'root@node-1:~#', and the user enters 'vim app.yaml', with the cursor positioned at the end of the command.

```
root@node-1:~# k create deploy nginx-app --image=nginx:1.11.10-alpine --dry-run=client -o y
aml > app.yaml
root@node-1:~# vim app.yaml
```

F:\Work\Data Entry Work\Data Entry\20200827\CKA\7 C.JPG

Readme Web Terminal THE LINUX FOUNDATION

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx-app
  template:
    metadata:
      labels:
        app: nginx-app
    spec:
      containers:
        - image: nginx:1.11.10-alpine
          name: nginx
~
~
~
~
~
~
~
'app.yaml
```

F:\Work\Data Entry Work\Data Entry\20200827\CKA\7 D.JPG

Readme Web Terminal THE LINUX FOUNDATION

```
root@node-1:~# k create deploy nginx-app --image=nginx:1.11.10-alpine --dry-run=client -o y
aml > app.yaml
root@node-1:~# vim app.yaml
root@node-1:~# k create -f app.yaml
deployment.apps/nginx-app created
root@node-1:~#
root@node-1:~#
root@node-1:~# k set image deploy nginx-app nginx=nginx:1.11.13-alpine --record
deployment.apps/nginx-app image updated
root@node-1:~# k rollout undo deploy nginx-app
deployment.apps/nginx-app rolled back
root@node-1:~#
```

Question: 75CKA: Actual Exam Q&A | **CLEARCATNET**

Create a pod with image nginx called nginx and allow traffic on port 80

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

kubectlrn nginx --image=nginx --restart=Never --port=80

Question: 76CKA: Actual Exam Q&A | **CLEARCATNET**

Create a namespace called 'development' and a pod with image nginx called nginx on this namespace.

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

kubectl create namespace development

kubectl run nginx --image=nginx --restart=Never -n development

Question: 78CKA: Actual Exam Q&A | **CLEARCATNET**

Create a Kubernetes secret as follows:

➤ Name: super-secretpassword: bob

❏ Create a pod named pod-secrets-via-file, using the redisImage, which mounts a secret named super-secretat /secrets.

Create a second pod named pod-secrets-via-env, using the redisImage, which exports password as CONFIDENTIAL

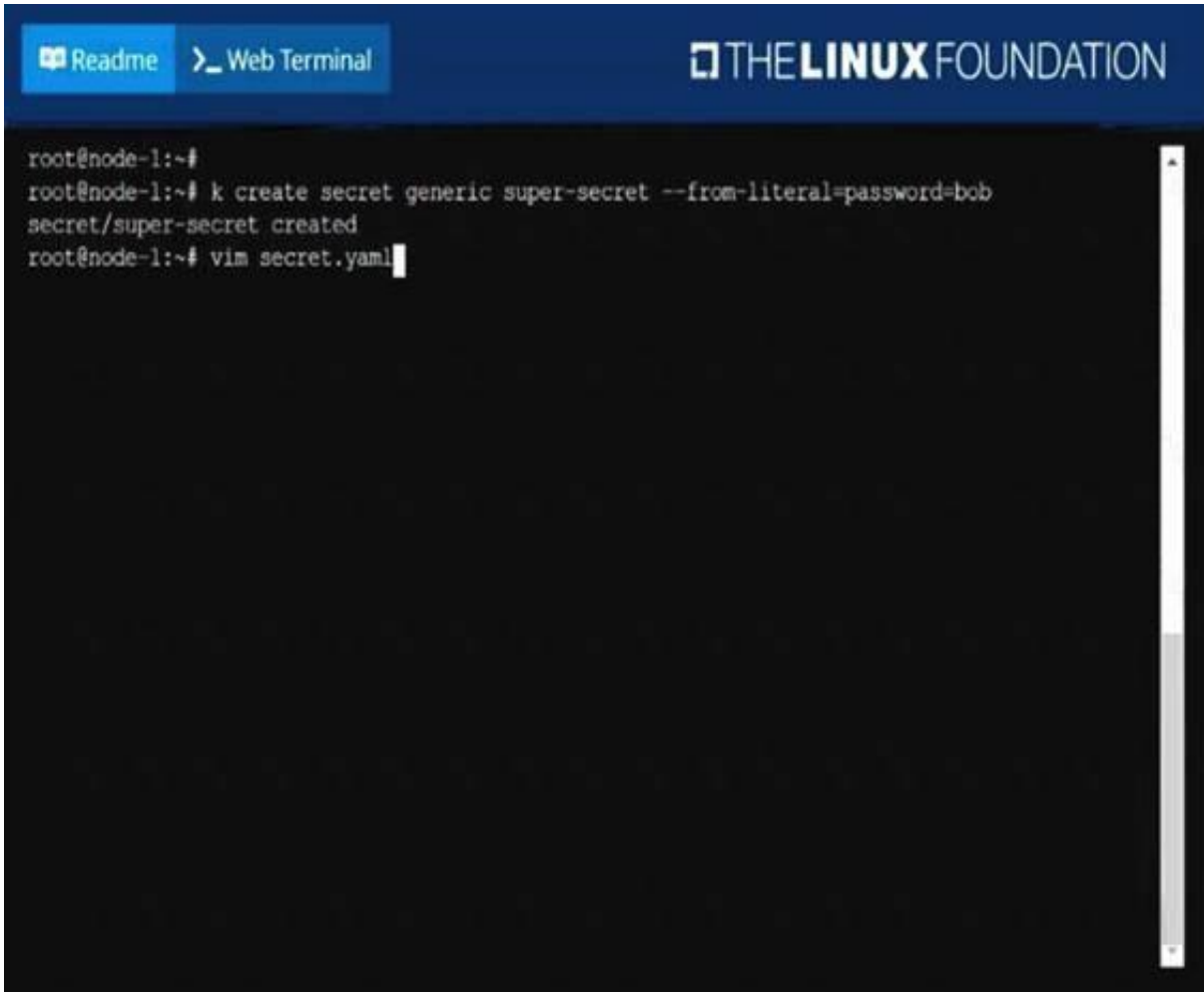
- A. Mastered
- B. Not Mastered

Answer: A

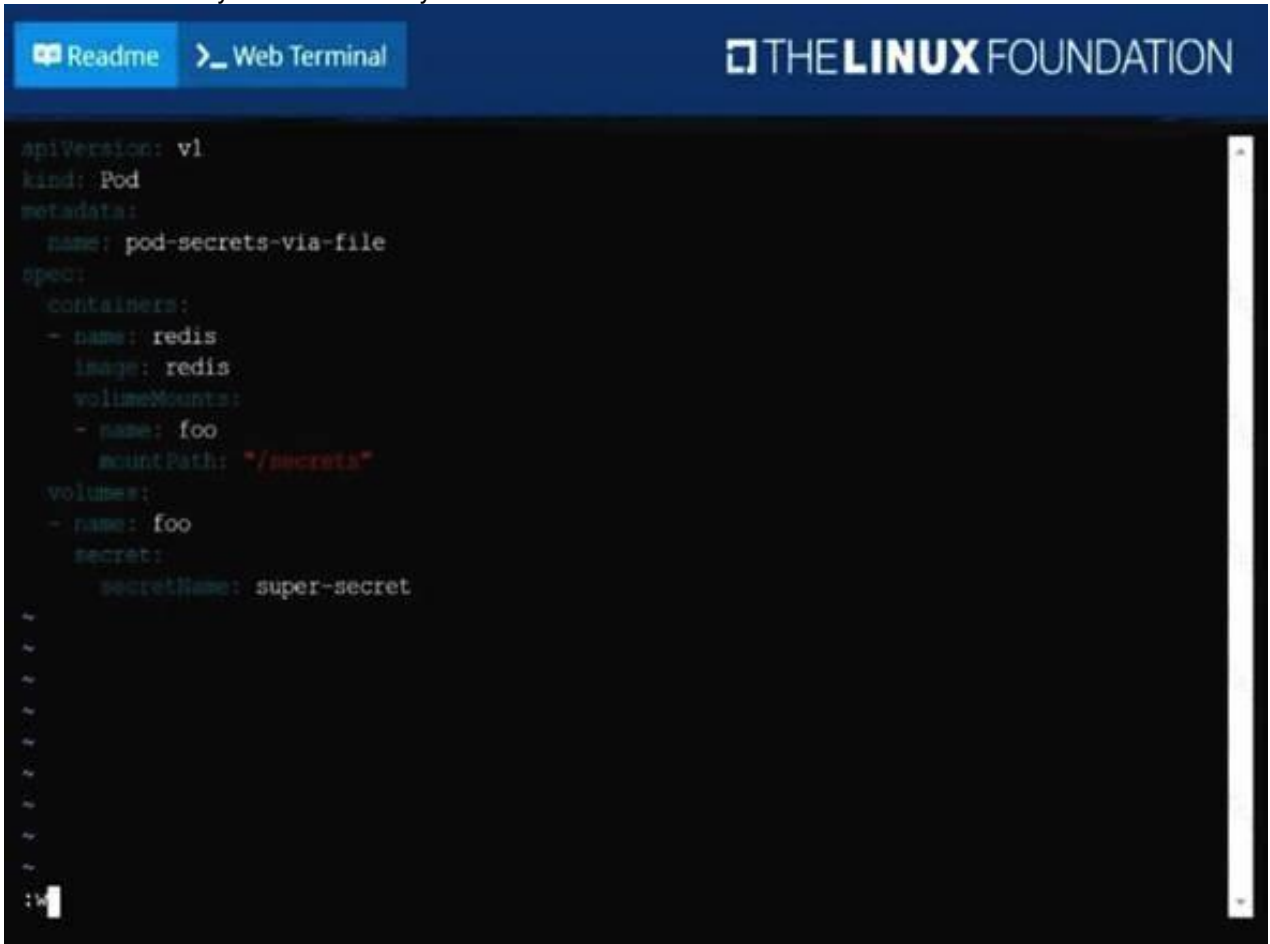
Explanation:

solution

F:\Work\Data Entry Work\Data Entry\20200827\CKA\12 B.JPG



F:\Work\Data Entry Work\Data Entry\20200827\CKA\12 C.JPG



F:\Work\Data Entry Work\Data Entry\20200827\CKA\12 D.JPG


```
root@node-1:~# k create -f secret.yaml
pod/pod-secrets-via-file created
root@node-1:~# vim secret1.yaml
root@node-1:~# k create -f secret1.yaml
pod/pod-secrets-via-env created
root@node-1:~# k get po
```

NAME	READY	STATUS	RESTARTS	AGE
cpu-utilizer-98b9se	1/1	Running	0	6h25m
cpu-utilizer-ab2d3s	1/1	Running	0	6h25m
cpu-utilizer-kipb9a	1/1	Running	0	6h25m
ds-kusc0020l-2r2k9	1/1	Running	0	40m
ds-kusc0020l-hzm9q	1/1	Running	0	40m
foo	1/1	Running	0	6h28m
front-end	1/1	Running	0	6h27m
hungry-bear	1/1	Running	0	36m
kucc8	3/3	Running	0	34m
nginx-app-848cfcf495-9prjh	1/1	Running	0	19m
nginx-app-848cfcf495-gl2kh	1/1	Running	0	19m
nginx-app-848cfcf495-pg2c8	1/1	Running	0	19m
nginx-kusc0010l	1/1	Running	0	26m
pod-secrets-via-env	1/1	Running	0	4s
pod-secrets-via-file	1/1	Running	0	106s
webserver-84c55967f4-qzjcv	1/1	Running	0	6h43m
webserver-84c55967f4-t479l	1/1	Running	0	6h43m

```
root@node-1:~#
```

Question: 79CKA: Actual Exam Q&A | **CLEARCATNET**

Perform the following tasks:

- Add an init container to hungry-bear (which has been defined in spec file /opt/KUCC00108/pod-spec-KUCC00108.yaml)
- The init container should create an empty file named /workdir/calm.txt. If /workdir/calm.txt is not detected, the pod should exit
- Once the spec file has been updated with the init container definition, the pod should be created

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

solution

F:\Work\Data Entry Work\Data Entry\20200827\CKA\4 B.JPG

```
root@node-1:~# vim ds.yaml
iroot@node-1:~# k create -f ds.yaml
daemonset.apps/ds-kusc00201 created
root@node-1:~# k get ds
NAME           DESIRED  CURRENT  READY  UP-TO-DATE  AVAILABLE  NODE SELECTOR  AGE
ds-kusc00201    2        2        2      2           2          <none>         4s
root@node-1:~# vim /opt/KUCC00108/pod-spec-KUCC00108.yaml
```

```
apiVersion: v1
kind: Pod
metadata:
  name: hungry-bear
spec:
  volumes:
    - name: workdir
      emptyDir: {}
  containers:
    - name: checker
      image: alpine
      command: ["/bin/sh", "-c", "if [ -f /workdir/calm.txt ]; then sleep 100000; else exit 1; fi"]
      volumeMounts:
        - name: workdir
          mountPath: /workdir
    initContainers:
      - name: create
        image: alpine
        command: ["/bin/sh", "-c", "touch /workdir/calm.txt"]
        volumeMounts:
          - name: workdir
            mountPath: /workdir
:~
```

F:\Work\Data Entry Work\Data Entry\20200827\CKA\4 D.JPG

Readme

Web Terminal

THE LINUX FOUNDATION

```
root@node-1:~# vim ds.yaml
iroot@node-1:~# k create -f ds.yaml
daemonset.apps/ds-kusc00201 created
root@node-1:~# k get ds
NAME          DESIRED  CURRENT  READY  UP-TO-DATE  AVAILABLE  NODE SELECTOR  AGE
ds-kusc00201   2        2        2      2           2          <none>         4s
root@node-1:~# vim /opt/KUCC00108/pod-spec-KUCC00108.yaml
root@node-1:~# k create -f /opt/KUCC00108/pod-spec-KUCC00108.yaml
pod/hungry-bear created
root@node-1:~#
```

Question: 79

CKA: Actual Exam Q&A | **CLEARCATNET**

Create and configure the service front-end-services so it's accessible through NodePort and routes to the existing pod named front-end.

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

solution
F:\Work\Data Entry Work\Data Entry\20200827\CKA\8 B.JPG

Readme Web Terminal THE LINUX FOUNDATION

```
root@node-1:~# k expose po
error: resource(s) were provided, but no name, label selector, or --all flag specified
See 'kubectl expose -h' for help and examples
root@node-1:~# k expose po fron-end --name=front-end-service --port=80 --target-port=80 --t
ype=NodePort
Error from server (NotFound): pods "fron-end" not found
root@node-1:~# k expose po front-end --name=front-end-service --port=80 --target-port=80 --
type=NodePort
service/front-end-service exposed
root@node-1:~# k get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)        AGE
front-end-service   NodePort    10.103.221.227 <none>         80:31828/TCP   3s
kubernetes          ClusterIP   10.96.0.1      <none>         443/TCP        77d
root@node-1:~#
```

Question: 80

CKA: Actual Exam Q&A | CLEARCATNET

Create an nginx pod and list the pod with different levels of verbosity

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

```
// create a pod
kubectl run nginx --image=nginx --restart=Never --port=80
// List the pod with different verbosity kubectl get po nginx --v=7 kubectl get po nginx --v=8 kubectl get po nginx --v=9
```

Question: 81

CKA: Actual Exam Q&A | CLEARCATNET

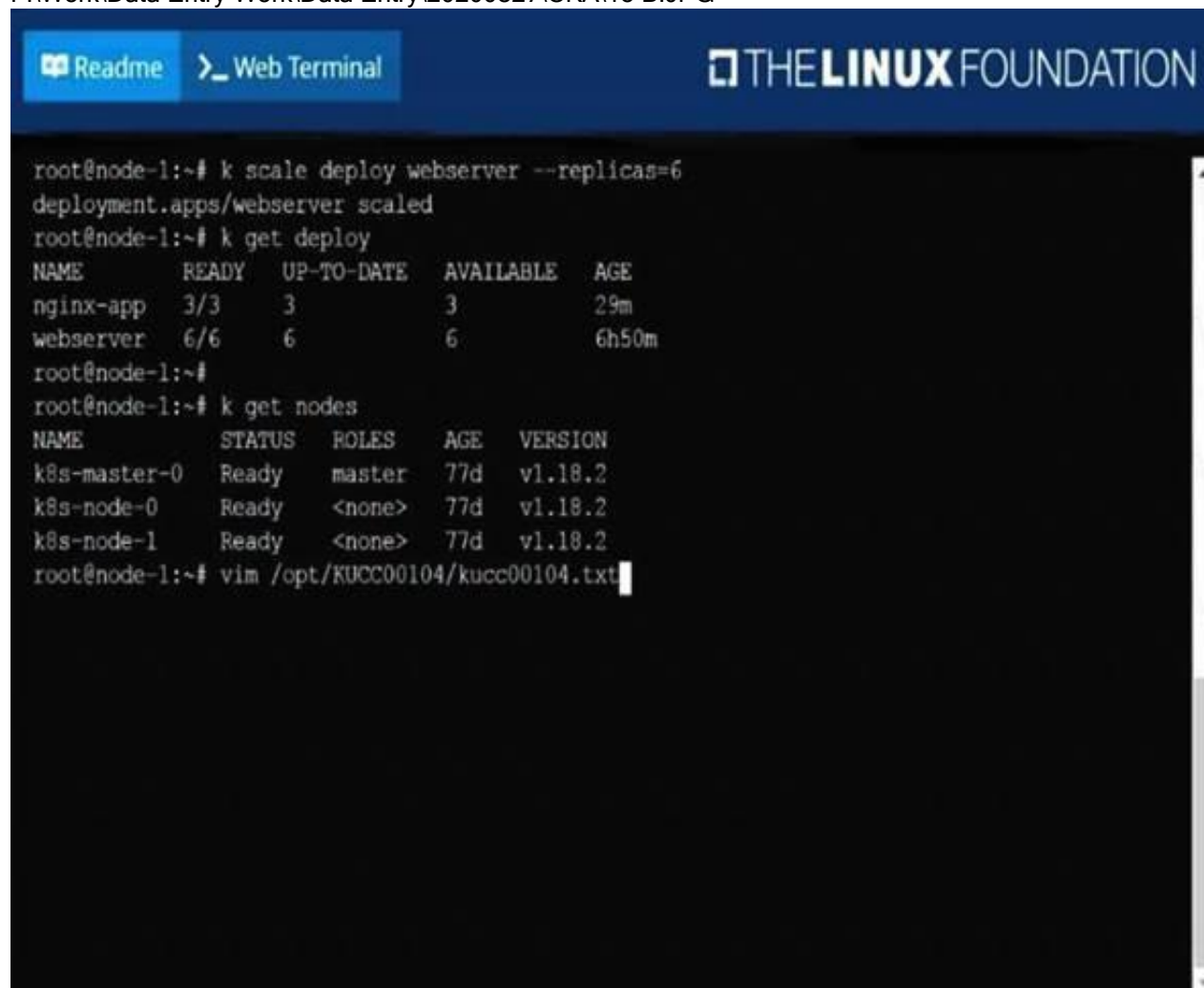
Check to see how many worker nodes are ready (not including nodes taintedNoSchedule) and write the number to/opt/KUCC00104/kucc00104.txt.

- A. Mastered
- B. Not Mastered

Answer: A

Explanation:

solution



The screenshot shows a web terminal interface with a dark blue header. On the left, there are two buttons: 'Readme' and 'Web Terminal'. On the right, the 'THE LINUX FOUNDATION' logo is displayed. The terminal window shows a series of commands and their outputs:

```
root@node-1:~# k scale deploy webserver --replicas=6
deployment.apps/webserver scaled
root@node-1:~# k get deploy
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
nginx-app     3/3     3            3           29m
webserver     6/6     6            6           6h50m
root@node-1:~#
root@node-1:~# k get nodes
NAME           STATUS   ROLES    AGE   VERSION
k8s-master-0   Ready   master   77d   v1.18.2
k8s-node-0     Ready   <none>    77d   v1.18.2
k8s-node-1     Ready   <none>    77d   v1.18.2
root@node-1:~# vim /opt/KUCC00104/kucc00104.txt
```

2

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

: WQ!

Best of Luck

You have worked hard to get to this point, and you are well-prepared for the exam. Keep your head up, stay positive, and go show that exam what you're made of!



VIST US to Purchase easily any IT Exam Dumps - WWW.CLEARCATNET.COM

STAY CONNECTED WITH US-

Send us your request/inquiry at clearcat.net@gmail.com or connect us for  [Live Support](#) any time for **any certification exam dumps pdf** Or for **most asked Interview Q&A PDFs** to ensure your success in first try!!



Follow us on: [Facebook](#) | [Instagram](#) | [LinkedIn](#) | [reddit](#) | [Twitter](#) | [Quora](#) | [YouTube](#)

All the best!!



Clearcat.net

Thank you!

 We provides all IT Exam dumps like AZURE **AWS**, **Google**, CISCO, **CompTIA** & many more vendor's dumps..

Send us your request/inquiry at clearcat.net@gmail.com any time for any certification exam dumps pdf Or for most asked Interview Q&A PDFs to ensure your success!!

Get Dumps PDF Now & Be Certified Quickly!

Subscribe us @Clearcatnet